

# QUBO: A Retrieval-Augmented Generation Powered Philippine Cultural History Chatbot

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**Abstract**—This study presents a Retrieval-Augmented Generation (RAG) system for Philippine cultural history question answering, integrating hybrid information retrieval with large language model generation to produce accurate, context-grounded responses. The proposed framework combines sparse keyword-based retrieval using BM25 with dense semantic retrieval and hybrid score fusion to efficiently identify relevant historical documents. A curated corpus of 34 documents derived from authoritative Philippine history sources was constructed, along with a gold-standard evaluation set of 198 manually annotated questions linked to reference answers and supporting evidence.

Experimental results demonstrate strong retrieval performance, achieving a Hit Rate of 91.50% and a Mean Reciprocal Rank (MRR) of 0.868 at an optimal retrieval depth of  $k = 3$ , indicating reliable context identification with minimal retrieval noise. Generation quality was evaluated using both automatic metrics and an LLM-as-a-Judge framework. The system attained a BERTScore of 0.86 and a ROUGE score of 0.56, reflecting strong semantic alignment and content coverage, while LLM-based evaluation using Qwen 2.5 yielded high scores for faithfulness (4.85), completeness (4.59), and correctness (4.44). These findings highlight the effectiveness of RAG-based architectures for historically grounded question answering and their potential for building scalable, reliable digital systems for cultural heritage and educational applications.

**Index Terms**—Retrieval-Augmented Generation, Semantic Chunking, Sparse Retrieval, Dense Retrieval, RAG Chatbot, Philippine Cultural History

## I. INTRODUCTION

The Philippines contains a broad cultural history which provides a significant part in developing its national identity, so historical education is an essential component of college curriculum [1], [2]. Readings in Philippine History and The Life and Works of Rizal are two courses that emphasize important analysis and interpretation, reflecting the dynamic and growing

nature of Philippine cultural history, which is impacted by different views and continuing studies [3]. As a result, learning tools in this discipline must address complexity, contextual accuracy, and interpretative variance.

At the same time, the fast adoption of digital technology in Philippine higher education has resulted in high student use of generative artificial intelligence tools. Recent research and publications show that the majority of Filipino college students use AI systems like ChatGPT for tasks such as text production, translation, concept explanation, and summarization, highlighting their expanding influence on academic study practices [4], [5]. However, general-purpose big language models are trained on huge, diverse datasets and are not always connected with curriculum-specific or institutionally verified information [6], [7]. This misalignment raises the possibility of delusions—likely but wrong outputs—which is particularly concerning in historically sensitive domains such as Philippine cultural history, where factual accuracy and source grounding are critical [8]–[11].

To address these difficulties, current research suggests that include outside sources into language model development procedures can increase accuracy and accountability [12]. Retrieval-Augmented Generation (RAG) meets this need by merging document retrieval with generative models, allowing answers based on selected, domain-specific sources without adjusting model parameters [13], [14]. While RAG has demonstrated potential in a variety of genres, its effectiveness is dependent on retrieval quality, document chunking methodologies, and evaluation methods, and it remains unexplored in the context of locally grounded Philippine historical content [15].

This study focuses into the application of Retrieval-Augmented Generation to assist with question responding

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based on Philippine cultural history books. The study's goal is to create a prototype chatbot that improves student engagement while maintaining historical accuracy and source alignment by analyzing various RAG setups and assessment techniques. The work adds to ongoing conversations about AI-assisted learning in higher education and promotes the preservation and transmission of Philippine cultural heritage.

## II. REVIEW OF RELATED LITERATURE AND STUDIES

### A. *Preservation and Accessibility of Philippine Cultural Heritage*

Educational institutions such as the NCCA, NHCP, and university libraries preserve the Filipino cultural heritage, which includes historical, literary, and social forms, and frequently provide expert commentary to assist education, research, and national identity [16], [17], [30], and [31]. Digital technology has proven critical for making these resources more accessible and maintaining fragile originals, as evidenced by platforms such as the Filipino Heritage Library, ASEAN Information Library, and Philippine eBook Hub [19], [32]–[34]. Despite these efforts, challenges such as inadequate materials, inappropriate categorization, environmental concerns, and limited preservation resources persist, emphasizing the need for additional steps to protect the Philippines' cultural legacy for future generations [35].

### B. *The Role of AI in Student Learning*

Artificial intelligence (AI), including machine learning and natural language processing, is transforming education by improving student learning, allowing individualized approaches, and increasing engagement and academic achievement [4], [5], [6], [13], [36], [38], [44], [51]. Artificial intelligence also helps educators by automating duties such as grading and attendance, providing data-driven insights into student performance, and allowing teachers to focus on higher-value instructional activities while changing in real time [38], [40], [49], and [52]. In the Philippine education system, AI provides opportunities for personalized learning and scalable solutions, but difficulties such as privacy, data security, and the digital divide require ethical regulations and cautious implementation [39], [44], [50], [86], [87].

### C. *Chatbots Powered by Large Language Models (LLMs)*

Chatbots provide on-demand learning help by providing fast responses, solving misunderstandings, and adapting answers to different student levels, making them especially helpful in different learning environments [36], [40], [52], [54]. Their ability to personalize responses has improved significantly since 2020, due to deep learning and large language models (LLMs), but they are still prone to hallucinations and are less effective at developing higher-order skills such as critical thinking [8], [13], [37], [53], [75]. To optimize benefits like as improved student performance, motivation, and reduced teacher workload, LLM-powered chatbots should be carefully blended with traditional teaching methods under proper settings [36], [40], [54].

### D. *Limitations of Large Language Models*

Large Language Models (LLMs) encounter issues such as hallucinations, lack of source attribution, outdated training data, and domain-specific biases that restrict their reliability in academic and knowledge-intensive tasks [8], [15], [18], [24], [64], [75], and [81]. These limitations are especially important in subjects such as Philippine cultural history, where reliable primary sources, and context-specific expertise are required [15], [24], [64], and [81]. Retrieval-Augmented Generation (RAG) approaches address these concerns by enabling LLMs to access validated external materials, so enhancing the accuracy of facts, transparency, and topic relevance [15], [24], and [63].

### E. *Information Retrieval*

Information retrieval (IR) is the study of how to store, retrieve, and get at huge quantities of organized or unstructured data, which serves as the foundation for tools such as search engines and digital libraries [55], [56]. Traditional methods for document ranking, such as TF-IDF and BM25, focus on statistical measurements, whereas contemporary approaches incorporate machine learning and NLP techniques, such as Sentence-BERT embeddings, to enable semantic search and better capture contextual relevance [40], [57], [59], and [60]. IR has become critical for supporting LLMs by giving correct and contextually appropriate information, which minimizes hallucinations and enhancing reliability in domains such as education [15], [61], and [62].

### F. *Retrieval-Augmented Generation*

Retrieval-Augmented generation (RAG) overcomes LLM shortcomings such as limited knowledge generalization and hallucinations by enabling models to obtain important information from external sources during text creation [15], [24], [63]. Instead of relying simply on pre-trained information, RAG enhances LLM outputs by searching for contextually relevant data, allowing for more accurate and customized replies to tasks like question answering, language recognition, and text completion [24], [65], [81]. In education, RAG guarantees that LLM responses are both relevant and verified, which improves dependability when interacting with learning materials and organizational knowledge bases [24], [81].

### G. *Advances in RAG*

Recent improvements in Retrieval-Augmented Generation (RAG) improve LLMs by resolving hallucinations, outdated information, and a lack of source reference, while leveraging retrieval techniques to anchor responses in real-world resources [24], [27], [59], [63]. Modern RAG systems use context processing, multi-stage retrieval, cross-document reasoning, and retrieval variety to provide accurate, relevant, and conceptually related information that supports difficult academic questions and deeper learning [61], [65], [66], [81], and [62]. These advancements improve RAG's basis for educational chatbots by enhancing retrieval accuracy, contextual comprehension, data security, and enabling course-specific advice for independent study [61], [62], [66], [81].

## H. Challenges and Limitations of RAG

Although Retrieval-Augmented Generation (RAG) improves LLM outputs by integrating external data, it faces significant challenges such as relying on the quality of retrieved documents, continued hallucinations, and high computational demands for embedding, indexing, and similarity search [1]–[4], [15], [24], [61], [62], [74]. Evaluation and benchmarking are still problematic due to a lack of defined measures, language biases towards high-resource datasets, and insufficient frameworks for distinguishing retrieval from creation mistakes [6], [63], [70]. Domain adaptability and transparency are also limited, particularly in culturally specialized domains such as Philippine history, where structured sources are few and source attribution is frequently insufficient, reducing reliability and confidence for educational purposes [66], [81].

## I. Rag applications in Student Learning

RAG-based systems increase accuracy and relevance by accessing current information from external knowledge bases, reducing the requirement for costly retraining [24], [63]. Studies on RAG academic chatbots, including as BiWi AI Tutor and the MARK system, demonstrate great factual accuracy and strong alignment with expert judgments, but they still struggle with reasoning tasks and rely heavily on the quality of their knowledge stores [67]–[69]. While these systems have the potential to provide scalable and reliable learning help in higher education, there are limitations in reasoning ability, personalization, and long-term school integration [67]–[69].

## III. METHODOLOGY

This section details the methodology used to develop the Retrieval-Augmented Generation (RAG) chatbot, Qubo. It covers the overall system architecture, the data collection process, the construction of the retrieval and generation pipeline, and the evaluation strategy employed to assess the system's performance.

### A. Overview of the RAG System

The RAG system architecture is composed of two main components: the document intake flow and the query flow, which operate together to prepare documents and generate responses. During the intake phase, documents are processed, segmented into smaller chunks, embedded into vector representations, and stored in a vector database for efficient similarity-based retrieval. The query phase embeds the user's question, retrieves and ranks the most relevant document chunks, and uses the top-ranked results as contextual input for a large language model to generate accurate responses.

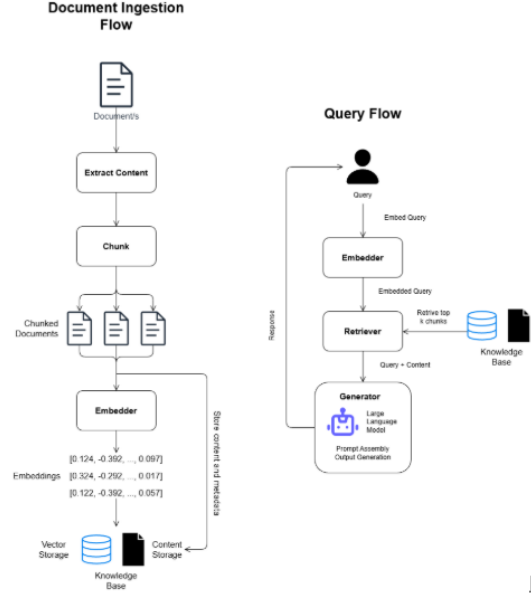


Fig. 1. Overview RAG System

### B. Content Extraction

The pipeline begins by converting source documents into machine-readable text. DOCX files are first converted into PDF format, after which content is extracted using pdfplumber, as it preserves document structure such as headers, tables, columns, and spacing. Compared to simpler text extraction approaches, this method minimizes errors and data misalignment, particularly in documents with complex layouts. For scanned or image-based documents, optical character recognition (OCR) is performed using Pytesseract to extract text embedded in images, ensuring that all content is processed consistently and that no essential information is overlooked.

### C. Chunking

Long texts are divided into chunks to improve retrieval accuracy; however, fixed-length chunking approaches often disrupt the natural flow of ideas by splitting related content. Semantic chunking addresses this issue by grouping text segments based on embedding similarity, allowing phrases with similar meanings to remain together. Overlap and token constraints are applied to preserve context and meet model limitations. This study adopts a hybrid two-stage approach using LangChain, where semantic segmentation is applied first to maintain logical coherence, followed by token-based splitting to manage chunk length.

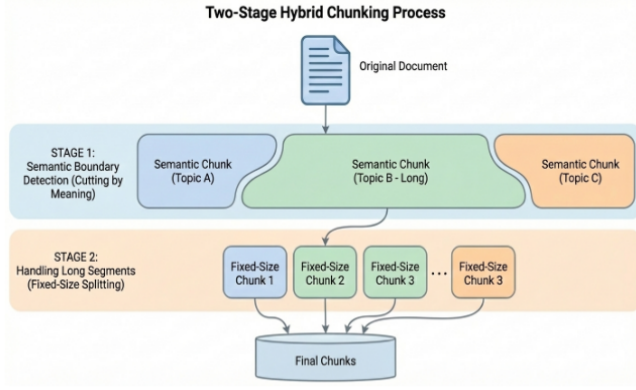


Fig. 2. Embedding Process

#### D. Embedding

Each text chunk is converted into an embedding, which is a numerical vector in a high-dimensional space that captures semantic meaning. These embeddings enable the retrieval engine to compare text chunks based on conceptual similarity rather than exact keyword matches, improving the relevance of retrieved results. The embedding process consists of three main stages.

- 1) **Input Preparation** – Each text chunk, typically consisting of several sentences, is provided as input to the embedding model.
- 2) **Vector Transformation** – The input text is transformed into a fixed-length numerical vector, where each dimension represents latent semantic features learned from large-scale language data.
- 3) **Semantic Space Mapping** – The resulting vectors are placed within a shared high-dimensional space, where semantically related chunks are positioned closer together even if different terminology is used. For instance, passages discussing “plants producing food” and “photosynthesis” are located near each other despite using different words.

This embedding strategy allows the retrieval system to rank text chunks based on their semantic similarity to a query, enabling accurate and context-aware information retrieval. MiniLM-L6 and Alibaba ModernBERT were evaluated to compare the performance of a lightweight, efficiency-focused model against a higher-capacity architecture designed for improved retrieval accuracy and extended context handling.

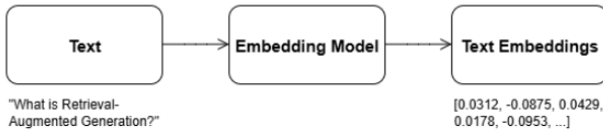


Fig. 3. Text Embedding

#### E. Vector Store

The embedded text chunks are stored in an FAISS vector database, which supports efficient similarity search over high-dimensional embeddings and enables large-scale retrieval with minimal latency. Cosine similarity is employed using L2 normalization and inner product search to ensure that similarity comparisons emphasize semantic alignment rather than vector magnitude. A flat index structure is used to guarantee accurate nearest-neighbor retrieval. LangChain’s document store maintains mappings between vector identifiers and their original text segments, allowing the system to reconstruct the appropriate context for the language model.

#### F. Query Embedding

Query embedding involves transforming a user’s natural language query into a numerical vector that represents its semantic meaning. By embedding both queries and documents within the same vector space, the system can retrieve relevant information based on semantic similarity rather than exact keyword matching. This approach enables more flexible and accurate retrieval.

#### G. Sparse Retrieval

Sparse retrieval represents documents and queries as high-dimensional sparse vectors and matches them using exact keyword overlap, which is efficiently supported by inverted indexing. Since simple word counts are insufficient for effective ranking, the BM25 algorithm is employed to score documents based on term frequency, inverse document frequency, and document length normalization. This approach provides an efficient, interpretable, and widely adopted method for keyword-based information retrieval. In this work, BM25 is used as the baseline sparse retrieval model.

$$\text{BM25}(D, Q) = \sum_{t \in Q} \text{IDF}(t) \cdot \frac{f(t, D) \cdot (k_1 + 1)}{f(t, D) + k_1 \cdot \left(1 - b + b \cdot \frac{|D|}{\text{avgdl}}\right)} \quad (1)$$

where:

- $D$  denotes a document,
- $Q$  denotes the query,
- $t$  denotes a term in the query,
- $f(t, D)$  is the frequency of term  $t$  in document  $D$ ,
- $|D|$  is the length of document  $D$  in words,
- $\text{avgdl}$  is the average document length in the collection,
- $k_1$  and  $b$  are tuning parameters, typically set to  $k_1 \in [1.2, 2.0]$  and  $b \approx 0.75$ .

The inverse document frequency (IDF) component of BM25 quantifies the importance of a term based on its distribution across the document collection. It is computed as:

$$\text{IDF}(t) = \ln \left( \frac{N - n_t + 0.5}{n_t + 0.5} + 1 \right), \quad (2)$$

where:

- $N$  is the total number of documents in the collection,
- $n_t$  is the number of documents containing term  $t$ .

This logarithmic formulation reduces the impact of extremely rare terms and stabilizes document scoring, thereby improving retrieval robustness and ranking quality.

#### H. Dense Retrieval

Dense vector search, which uses deep learning models like BERT, represents queries and documents as high-dimensional embeddings, allowing for semantic retrieval rather than exact keyword matching. Documents are retrieved by calculating the similarity between query and document embeddings in a continuous vector space using metrics like dot product and cosine similarity. Cosine similarity is especially useful since it compares vector orientation rather than size, allowing for accurate detection of semantic importance in high-dimensional data.

$$\text{CosineSimilarity}(A, B) = \frac{A \cdot B}{\|A\| \|B\|} \quad (3)$$

where:

- $A \cdot B$  denotes the dot product of vectors  $A$  and  $B$ ,
- $\|A\|$  and  $\|B\|$  denote the Euclidean norms (magnitudes) of vectors  $A$  and  $B$ .

The resulting similarity score ranges from  $-1$  to  $1$ , where a value of  $1$  indicates perfect alignment between the vectors, implying maximum similarity; a value of  $0$  indicates orthogonality, corresponding to no similarity; and a value of  $-1$  indicates complete opposition.

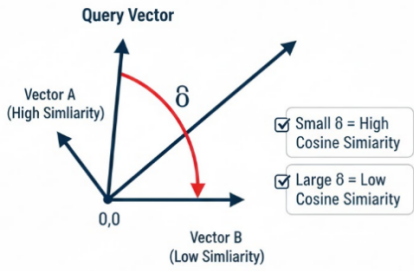


Figure 7. Cosine Similarity

Fig. 4. Cosine Similarity

#### I. Hybrid Retrieval

Hybrid retrieval combines sparse and dense retrieval techniques to improve both exact keyword matching and semantic understanding. It employs a weighted fusion strategy to combine relevance scores from methods like BM25 and cosine similarity into a single final score. This method increases retrieval precision as well as durability by prioritizing documents that are both literally and semantically relevant.

$$\text{FusedScore}(d) = \alpha \cdot s_{\text{dense}}(d) + (1 - \alpha) \cdot s_{\text{BM25}}(d) \quad (4)$$

where:

- $d$  is a candidate document,
- $s_{\text{dense}}(d)$  is the similarity score of document  $d$  produced by the dense retriever (e.g., cosine similarity between embedding vectors),
- $s_{\text{BM25}}(d)$  is the relevance score of document  $d$  produced by the sparse retriever using BM25,
- $\alpha \in [0, 1]$  is a weighting factor that controls the relative contribution of dense and sparse retrieval scores.

#### J. Reranker

After retrieving candidate documents, we apply a reranking stage using a pretrained cross-encoder model named BAAI/bge-reranker-base a cross-encoder reranker model developed by the Beijing Academy of Artificial Intelligence (BAAI). It belongs to the BGE (BAAI General Embedding) model family, designed for retrieval and reranking tasks in information retrieval (IR) and Retrieval-Augmented Generation (RAG) pipelines [106]. Unlike bi-encoder used in dense retrievers, that transforms a query to find relevant documents into a vector space [107], that independently encode queries and documents into embeddings for similarity comparison in vector space, a cross-encoder jointly processes each query-document pair and produces a direct relevance score (logit) [108]. It then computes a relevance score that indicates how well the passage matches the query. This joint encoding allows the model to capture richer semantic interactions between the query and document text, which improves retrieval precision compared to bi-encoder-based approaches [3], [4].

The output of the BAAI/bge-reranker-base cross encoder is unbounded. To make easy to interpret scores we applied a sigmoid transformation to map raw logits into a probabilistic like  $[0, 1]$  range:

$$s(d | q) = \sigma(\text{CE}(q, d)) = \frac{1}{1 + e^{-\text{CE}(q, d)}} \quad (5)$$

By running a cross encoder to the query-passage pair, we ensured semantic relevance and precision. We then sent the top 10 documents from the Cross encoder reranked as context to the generation module.

1) *Model Selection*: This study adopts a model selection phase to evaluate the suitability of different pretrained Large Language Models (LLMs) for response generation within the Retrieval-Augmented Generation (RAG) framework. Rather than relying on a single model, multiple instruction-tuned LLMs were considered to examine trade-offs in response quality, computational efficiency, and instruction-following capability.

2) *Prompt Assembly*: Prompt assembly refers to the careful construction of the input provided to the LLM to ensure accurate, contextually grounded, and instruction-aligned responses. The prompt combines the user's query, retrieved context chunks, and explicit instructions to guide the model's behavior as seen in Figure 7. The system prompt instructs the model to use only the information present in the context, avoid inventing facts or citations, and respond with a standard message if the answer cannot be found. Few-shot examples are



included to illustrate the expected format and citation style. This structured assembly minimizes hallucinations, enforces factual accuracy, and ensures that the model produces concise, academically appropriate answers.

#### IV. RESULTS AND DISCUSSION

##### A. Experimental Setup

The experiments evaluated multiple configurations of a Retrieval-Augmented Generation (RAG) system for Philippine cultural history question answering in a controlled environment using an Intel i5 (10th gen) CPU and NVIDIA RTX 3060 GPU, implemented in Python with LangChain and Together AI-hosted LLMs. The dataset consisted of 34 curated documents derived from a Philippine history source book, with a gold-standard evaluation set of 198 questions manually created and linked to reference answers and evidence passages. Documents were preprocessed using a two-stage hybrid chunking strategy combining semantic segmentation with token-based splitting to balance semantic coherence and context window constraints. System evaluation was conducted in two stages: retrieval performance was measured using Hit Rate and MRR at varying  $k$  values, while generation quality was assessed using automatic metrics (BLEU, ROUGE, BERTScore) and an LLM-as-Judge framework evaluating correctness, faithfulness, and completeness.

##### B. Retrieval Results

Table IX presents the quantitative performance of the hybrid retrieval system across varying top- $k$  depths. The system achieved a Hit Rate of 83.00% and an MRR of 0.830 at  $k=1$ . This indicates that for most queries, the most relevant document is correctly identified as the very first result, validating the efficacy of the BM25 sparse. Expanding the retrieval scope to  $k=3$  improves the Hit Rate to 91.50%, with a corresponding MRR increase to 0.868. The alignment between the Hit Rate and MRR scores confirms that even when the correct answer is not at rank 1, it consistently appears near the top of the list, rather than being buried deep in the retrieval window.

TABLE I  
RETRIEVAL PERFORMANCE METRICS AT VARYING  $k$  VALUES

| Metric       | k=1   | k=3   | k=5   | k=10  |
|--------------|-------|-------|-------|-------|
| Hit Rate (%) | 83.00 | 91.50 | 93.20 | 94.10 |
| MRR          | 0.830 | 0.868 | 0.872 | 0.874 |

Figure 1 visually illustrates the performance trajectory, revealing a distinct "elbow point" at  $k=3$  where the system reaches optimal efficiency. The curve shows a steep ascent from  $k=1$  to  $k=3$ , representing a significant performance gain of 8.5%. However, beyond this point, the curve flattens into a plateau of diminishing returns. Increasing the retrieval depth from  $k=3$  to  $k=5$  yields only a marginal improvement of 1.7% (91.50% to 93.20%). This saturation pattern suggests that retrieving more than three documents adds minimal informational value while potentially introducing irrelevant noise. Consequently,  $k=3$  is selected as the optimal cutoff for the

generation phase, balancing high recall with context window efficiency.

##### C. Generation Results

The generation quality was evaluated using the optimal retrieval setting  $k=3$ , using it into the Meta-Llama-3-70B Instruct Turbo. The model was tasked with answering queries using the exact same prompt configuration shown in figure 1.

```

Answer the question using ONLY the provided context.
If the answer is not present, say:
"The information is not available in the provided documents."

Context:
{context}

Question:
{query}

Answer:
===

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Fig. 5. Large Language Prompt

##### D. Automatic Metrics Result

The RAG system using the Meta Llama model showed strong performance, achieving a BERT Score of 0.86, which indicates that the generated responses closely matched the intended meaning of the reference answers. Its ROUGE score of 0.56 suggests that most of the important information and key details were successfully retained during generation. Although the BLEU score of 29.28 was lower, this mainly reflects differences in wording and sentence structure, showing that the model prioritizes conveying accurate meaning rather than copying text verbatim.

TABLE II  
AUTOMATIC EVALUATION METRICS

| Metric     | Score | Range   | Interpretation                                         |
|------------|-------|---------|--------------------------------------------------------|
| BERT Score | 0.86  | 0 – 1   | High semantic similarity; captures meaning accurately. |
| ROUGE      | 0.56  | 0 – 1   | Strong content retrieval and coverage.                 |
| BLEU       | 29.28 | 0 – 100 | Natural phrasing (paraphrased rather than copied).     |

To evaluate the semantic quality of the generated responses, an LLM-as-a-Judge framework based on Qwen 2.5 was employed. The model assessed the outputs of the Retrieval V4 + Meta Llama (Llama-3-70B) configuration across three dimensions: correctness, faithfulness, and completeness. The system achieved high scores in all criteria, with faithfulness obtaining the highest score of 4.85, indicating strong adherence to the retrieved context and minimal hallucination. Completeness scored 4.59, reflecting comprehensive coverage of the user queries, while correctness achieved 4.44, confirming a high degree of factual accuracy. Overall, these results demonstrate

the system’s effectiveness in synthesizing retrieved information to generate reliable and comprehensive answers.

TABLE III  
LLM-AS-A-JUDGE EVALUATION RESULTS

| Evaluation Metric | Score (1–5) | Interpretation                   |
|-------------------|-------------|----------------------------------|
| Faithfulness      | 4.85        | High adherence to source context |
| Completeness      | 4.59        | Comprehensive coverage of query  |
| Correctness       | 4.44        | Factually accurate response      |

### E. Conclusion

This study presented a Retrieval-Augmented Generation (RAG) system for Philippine cultural history question answering, integrating sparse and dense retrieval strategies with large language model generation. The experimental results demonstrate that the proposed hybrid retrieval framework effectively identifies relevant historical context, achieving a Hit Rate of 91.50% and an MRR of 0.868 at the optimal retrieval depth of  $k = 3$ . These results confirm that the system reliably retrieves pertinent documents while maintaining computational efficiency and minimizing contextual noise.

Generation quality evaluation further validates the effectiveness of the approach. Automatic metrics revealed strong semantic alignment and content coverage, with a BERTScore of 0.86 and ROUGE score of 0.56, indicating that the generated responses accurately capture the intended meaning and essential details of the reference answers. Moreover, the LLM-as-a-Judge evaluation based on Qwen 2.5 yielded high scores across faithfulness (4.85), completeness (4.59), and correctness (4.44), demonstrating the system’s ability to produce factually accurate, context-grounded, and comprehensive responses. Collectively, these findings highlight the potential of RAG-based architectures to support reliable and scalable question answering systems for culturally rich and historically grounded domains.

Future research may explore several directions to further enhance the performance and applicability of the proposed RAG system. First, incorporating more advanced dense retrieval models and domain-adapted embeddings could improve semantic matching, particularly for complex and context-dependent historical queries. Second, adaptive retrieval strategies that dynamically adjust the value of  $k$  based on query complexity could optimize the balance between recall and efficiency.

Expanding the dataset to include a broader range of Philippine historical sources, such as archival documents, oral histories, and multilingual materials, may significantly improve system coverage and robustness. Additionally, integrating multimodal data, including historical images, maps, and timelines, could enable richer and more interactive question answering experiences.

Finally, future work may focus on developing human-in-the-loop evaluation protocols and fine-tuning generation models using reinforcement learning from human feedback (RLHF) to further enhance answer quality, factual reliability, and interpretability. These advancements would contribute toward

building a comprehensive and trustworthy digital assistant for Philippine cultural heritage exploration.

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